MIL-M-28787/229A(EC)
12 October 1983
SUPERSEDING
MIL-M-28787/229(NAVY)
9 May 1975

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

AMPLIFIER, SUMMING, ANALOG, KEY CODE BEE

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic summing amplifier module.

* 1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.959 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be BEE.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 20 contact pins positioned A21 through A40. Contact pins A1 through A20 are optional.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/229-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

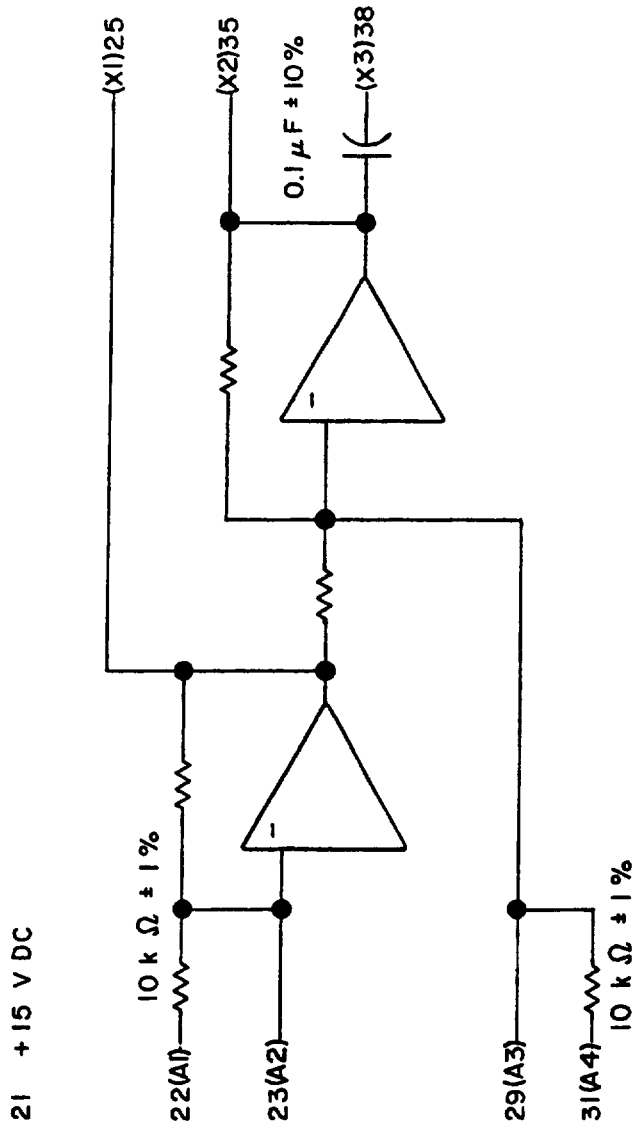
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTE: Pins 24, 26, 27, 28, 32, 33, 34, 36, 37, and 39 (and optional pins 1 thru 20) are unused.

FIGURE 1. BEE module function diagram, summing amplifier.

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*TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1,A2	Isolation	II	S	10	∞	1	∞	10	∞	1	∞	MΩ
B	Output short circuit current duration	III	Q	1	-	1	-	1	-	1	-	Minute
C1,C2	Input resistance	IV	Q	9.9	10.1	9.8	10.2	9.8	10.2	9.8	10.2	kΩ
D1-D4	DC gain accuracy check	V	X	9.7	10.3	9.7	10.3	9.7	10.3	9.7	10.3	V
D5	AC gain accuracy check	V	X	0.95	1.05	0.95	1.05	0.95	1.05	0.95	1.05	V rms
D6-D11	AC gain accuracy check	V	X	0.97	1.03	0.97	1.03	0.97	1.03	0.97	1.03	V rms
E	Output capacitance	VI	Q	0.09	0.11	0.09	0.11	0.08	0.12	0.08	0.12	µF
F1,F2	Output offset voltage	VII	X	-20	20	-20	20	-20	20	-20	20	mV
G1,G2	Output noise	VIII	S	0	100	0	100	0	100	0	100	µV rms
H1,H2	Quiescent power supply current	IX	Q	1	20	1	20	1	20	1	20	mA
J1,J2	Power supply transient rise time	X	Q	25	75	25	75	25	75	25	75	µs
K1-K4	Power supply ripple rejection check	XI	S	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	mV rms
L1	Large signal bandwidth check	XII	X	2.90	3.10	2.90	3.10	2.90	3.10	2.90	3.10	V rms
L2,L4	Large signal bandwidth check	XII	X	0	6	0	6	0	6	0	6	V rms
L3	Large signal bandwidth check	XII	X	0.0	3.1	0.0	3.1	0.0	3.1	0.0	3.1	V rms
M1-M4	Phase shift	XIII	X	178	182	178	182	178	182	178	182	Degree
M5,M6	Phase shift	XIII	X	0.2	4.2	0.2	4.2	0.2	4.2	0.2	4.2	Degree
M7,M8	Phase shift	XIII	X	12.8	16.8	12.8	16.8	11.9	20.0	11.9	20.0	Degree
N1,N2	Output voltage swing	XIII	S	10.00	15.15	10.00	15.15	10.00	15.15	10.00	15.15	Vpk

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through N shall be performed in any order with the following exceptions. If test conditions A or B are part of a sequence of tests, the order of testing shall be as follows: A, B where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through N in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through N shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. The ammeters XO1-1 and XO1-2 shall be monitored during the test and the readings shall not deviate more than ±10 percent from the initial current readings taken at maximum class temperature at the start of the test.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition H, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Mechanical shock, thermal shock, humidity, salt fog, and durability tests. The environmental tests shall be conducted in accordance with MIL-M-28787. Following the environmental tests, the electrical tests of table I which are marked with an (X) in the test type column, and test condition N, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.4 Life test. The life test shall be conducted in accordance with MIL-M-28787 except that the module shall be operated for a minimum of 1,000 hours. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C through N shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through N shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Military specification MIL-M-28787/229 supersedes Department of the Navy, Naval Electronic Systems Command Drawing No. 28687-0102-875. Part number M28787/229-1 supersedes part numbers 0102-875 and M28787/229.

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* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I.

Supply voltage	+15 V dc ±10% -15 V dc ±10%
Quiescent supply current (each supply)	20 mA maximum
Maximum input signal	7 V rms
Output voltage swing (R_L = 2 kΩ, f = 20 kHz)	10 Vpk minimum
Input resistance	10 kΩ ±1%
Gain	Unity
Bandwidth	DC to greater than 100 kHz

* 6.5 Module function. This module contains two unity gain inverting amplifiers in cascade. Both outputs are accessible and optional inputs are provided to allow different gains to be programmed into the amplifiers. The final amplifier has two outputs to allow either capacitive coupling or direct coupling, as required.

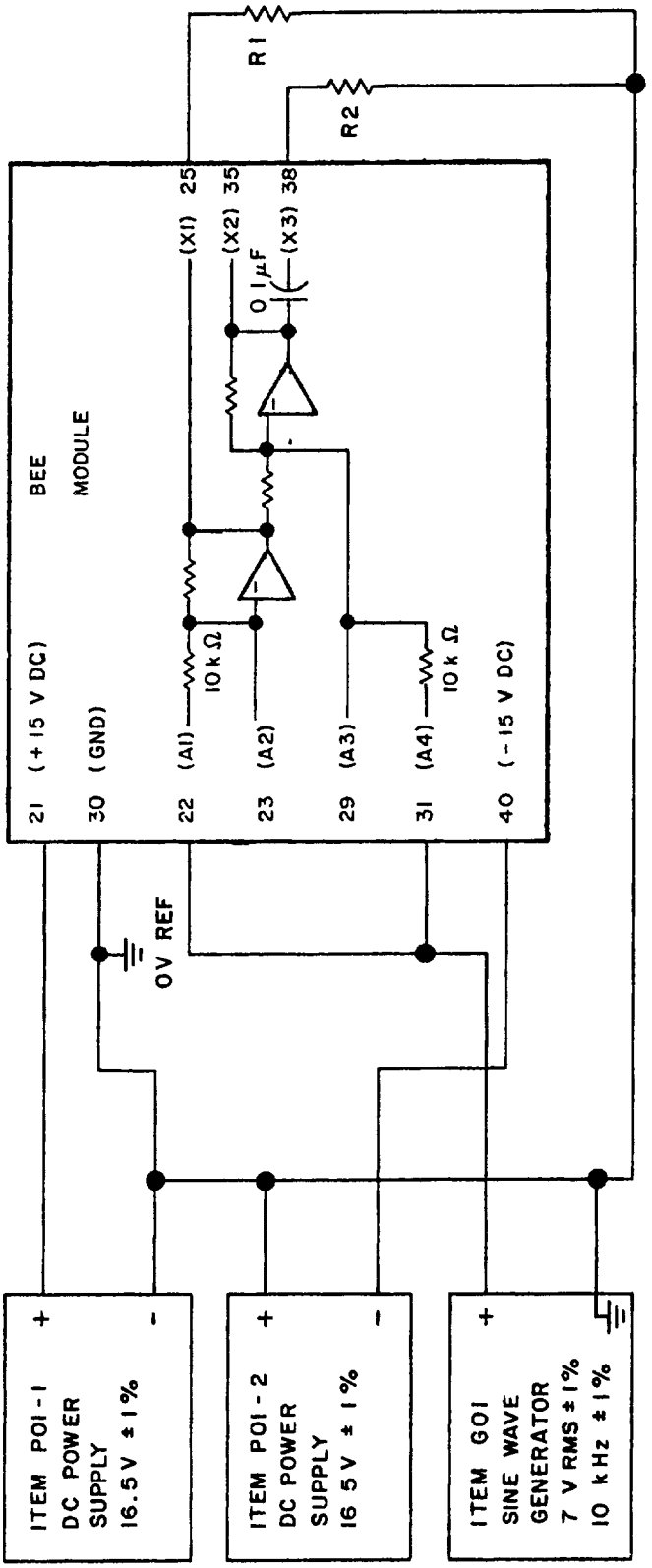
6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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NOTE: R1 = R2 = 2 kΩ ±1%, 1/8 W.

FIGURE 2. Operating temperature and life test circuit.

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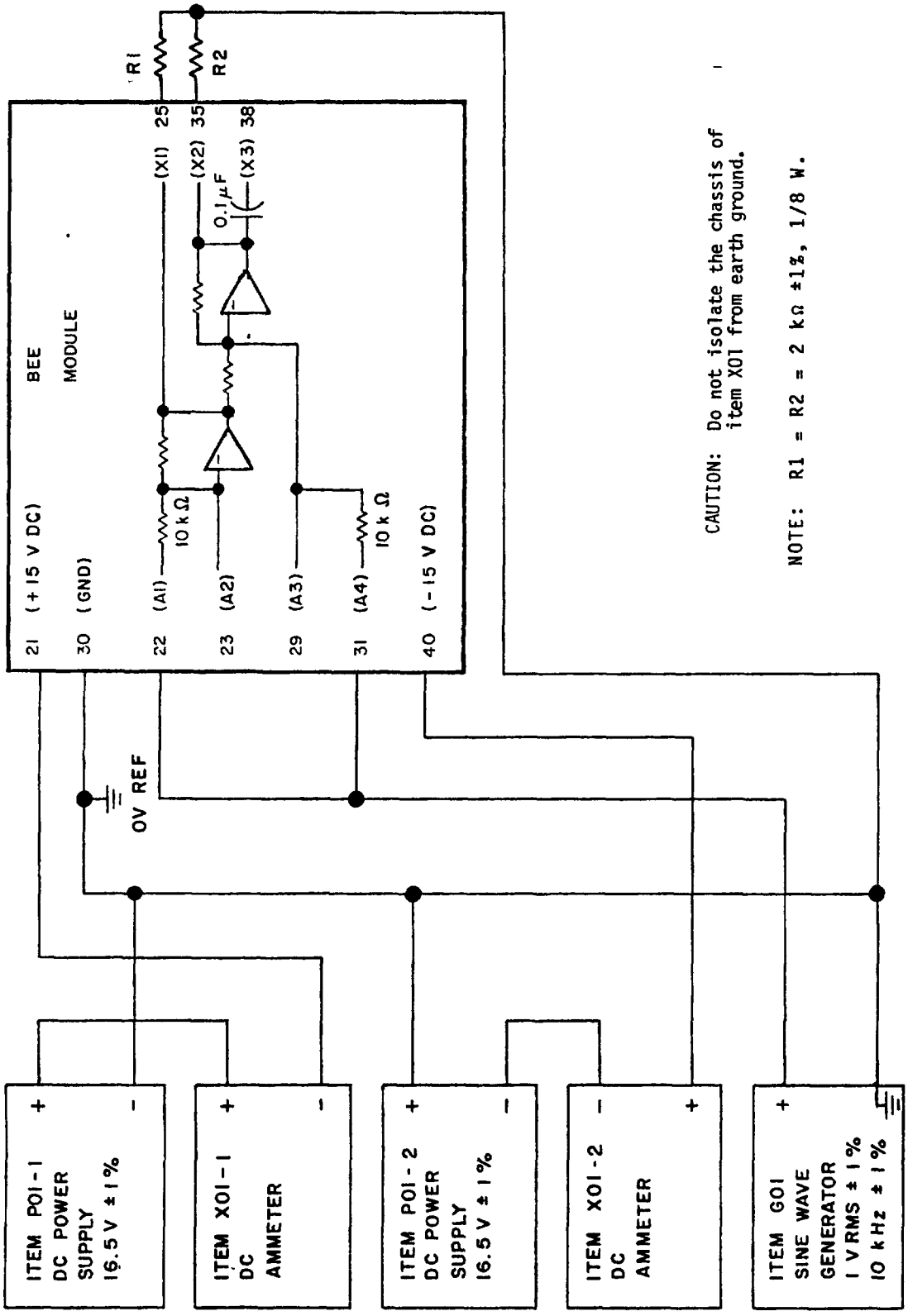


FIGURE 3. Component temperature test circuit.

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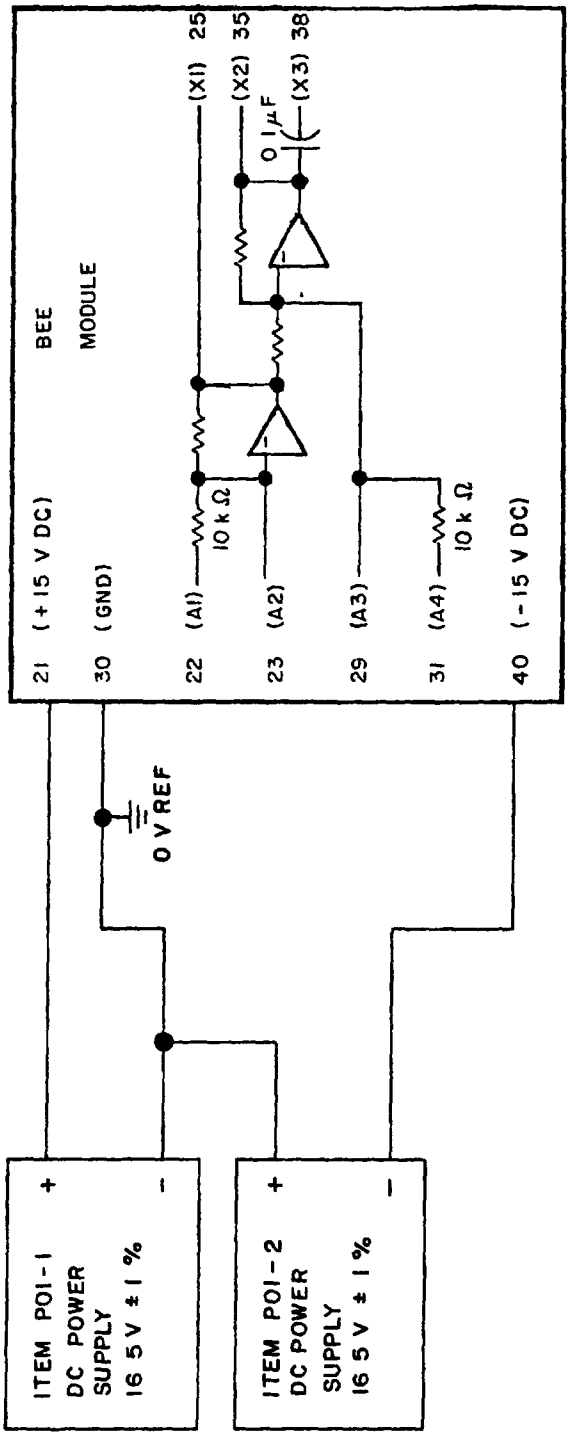


FIGURE 4. Vibration test circuit.

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*TABLE II. Isolation test.

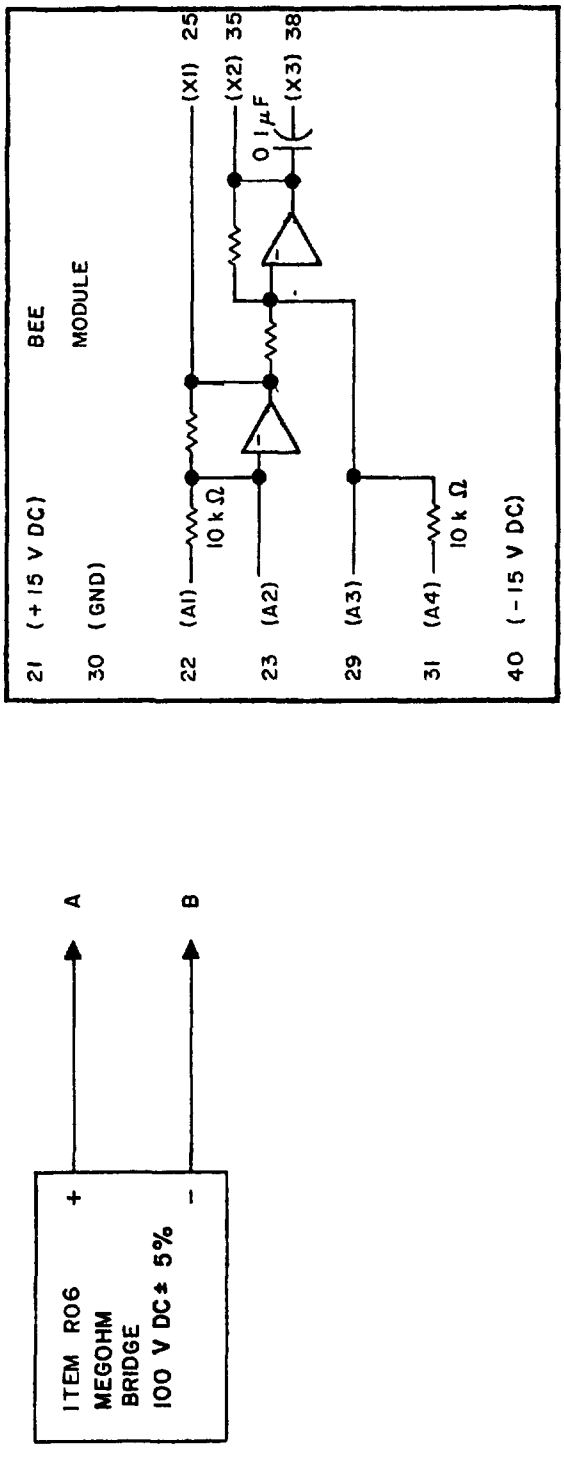
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation, frame	All pins	Frame	10	∞	MΩ	5
A2	Isolation, unused pins	All other pins	24,26,27,28,32, 33,34,36,37,39	10	∞	MΩ	5

1/ If the optional 20 pins are supplied (see 3.5.4), they shall be included under "A to pin" for test condition A1 and "B to pin" for test condition A2.

*TABLE III. Output short circuit current duration test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
B	Leave output shorted for 1 minute. Module shall pass the remainder of the tests as proof of no damage.	1	-	Minute	6

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WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: For test condition A1, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 5. Isolation test circuit for table II.

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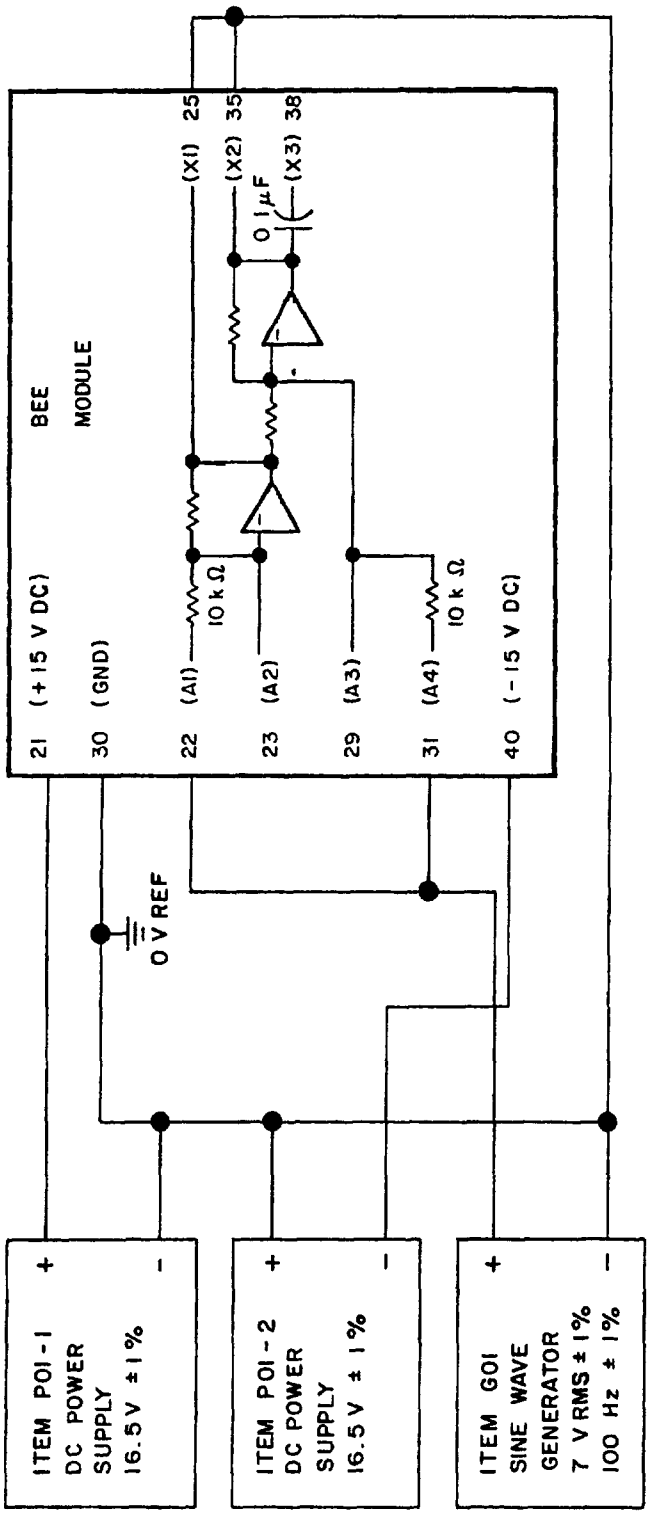


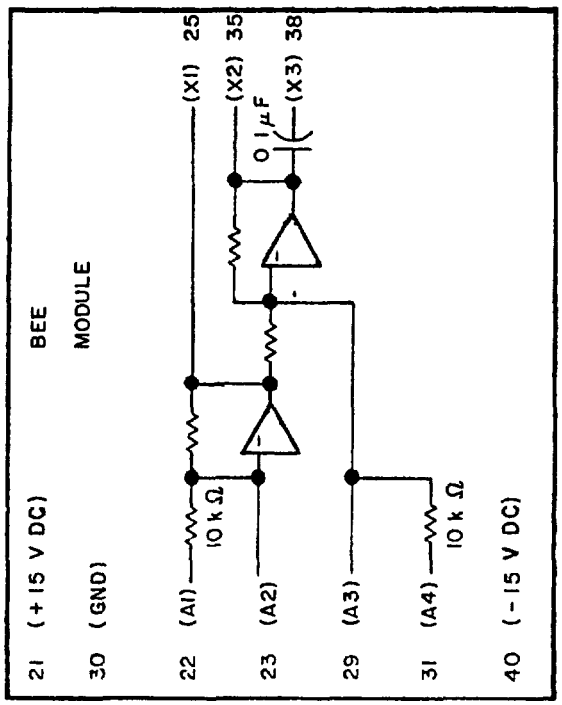
FIGURE 6. Output short circuit current durahon test circuit for table III.

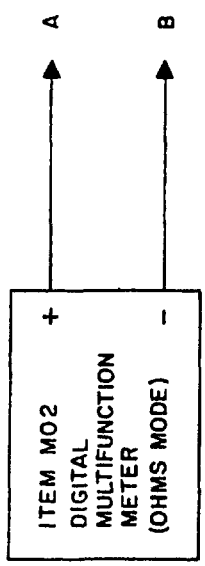
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TABLE IV. Input resistance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	A1 Input resistance	22	23	9.9	10.1	kΩ	7
C2	A4 Input resistance	29	31	9.9	10.1	kΩ	7

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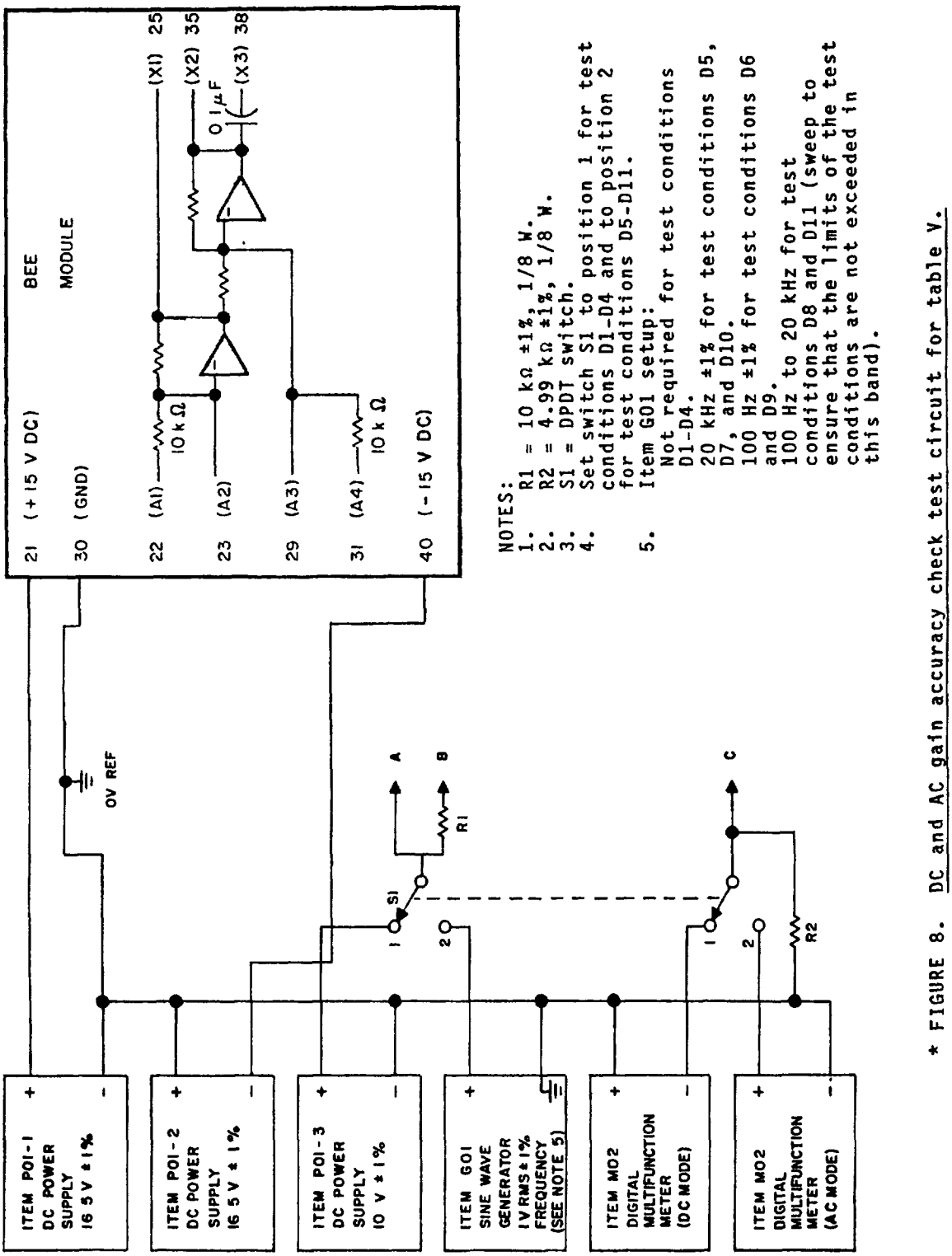
* FIGURE 7. Input resistance test circuit for table IV.

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TABLE V. DC and AC gain accuracy check test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
D1	X1 DC gain accuracy check	22	Open	25	9.7	10.3	V	8
D2	X1 "	Open	23	25	"	"	"	"
D3	X2 "	31	Open	35	"	"	"	"
D4	X2 "	Open	29	"	"	"	"	"
D5	X2 AC gain accuracy check	22	Open	"	0.95	1.05	V rms	"
D6	X1 "	"	"	25	0.97	1.03	"	"
D7	X1 "	"	"	"	"	"	"	"
D8	X1 "	"	"	"	"	"	"	"
D9	X2 "	31	"	35	"	"	"	"
D10	X2 "	"	"	"	"	"	"	"
D11	X2 "	"	"	"	"	"	"	"

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NOTES:
1. R1 = 10 kΩ ±1%, 1/8 W.
2. R2 = 4.99 kΩ ±1%, 1/8 W.
3. S1 = DPDT switch.
4. Set switch S1 to position 1 for test conditions D1-D4 and to position 2 for test conditions D5-D11.
5. Item G01 setup:
   Not required for test conditions D1-D4.
   20 kHz ±1% for test conditions D5, D7, and D10.
   100 Hz ±1% for test conditions D6 and D9.
   100 Hz to 20 kHz for test conditions D8 and D11 (sweep to ensure that the limits of the test conditions are not exceeded in this band).

* FIGURE 8. DC and AC gain accuracy check test circuit for table V.

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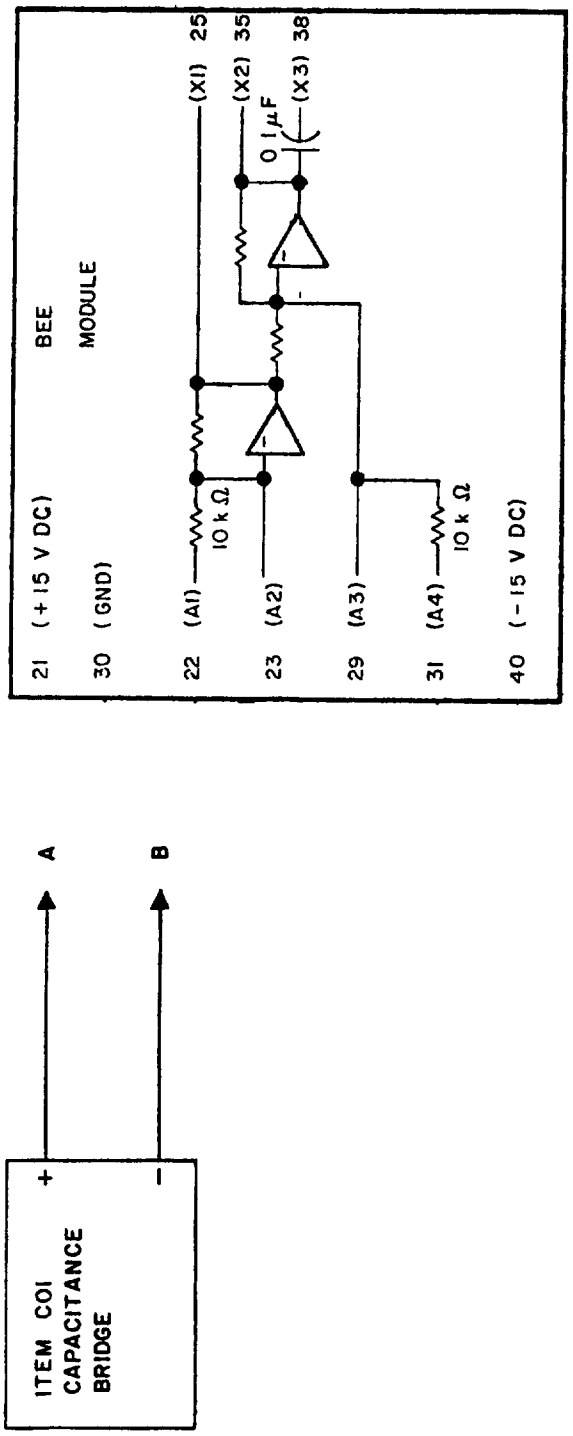
TABLE VI. Output capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E	X3 Output capacitance	38	35	0.09	0.11	µF	9

TABLE VII. Output offset voltage test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F1	X1 Output offset voltage	25	Open	30	-20	20	mV	10
F2	X2 Output offset voltage	35	30	Open	-20	20	mV	10

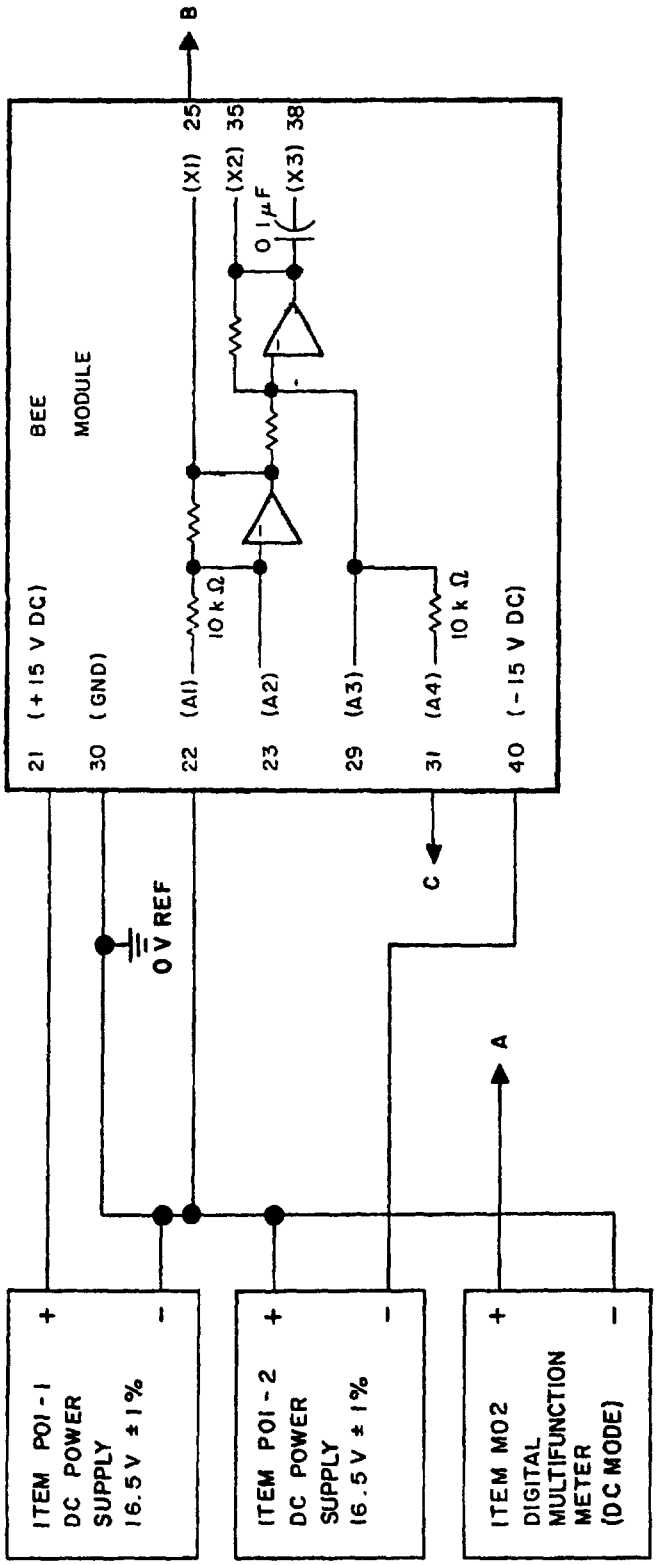
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NOTE: Item C01 setup:
Frequency = 10 kHz.
V_{pk} = 0.3 V maximum.

* FIGURE 9. Output capacitance test circuit for table VI.

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* FIGURE 10. Output offset voltage test circuit for table VII.

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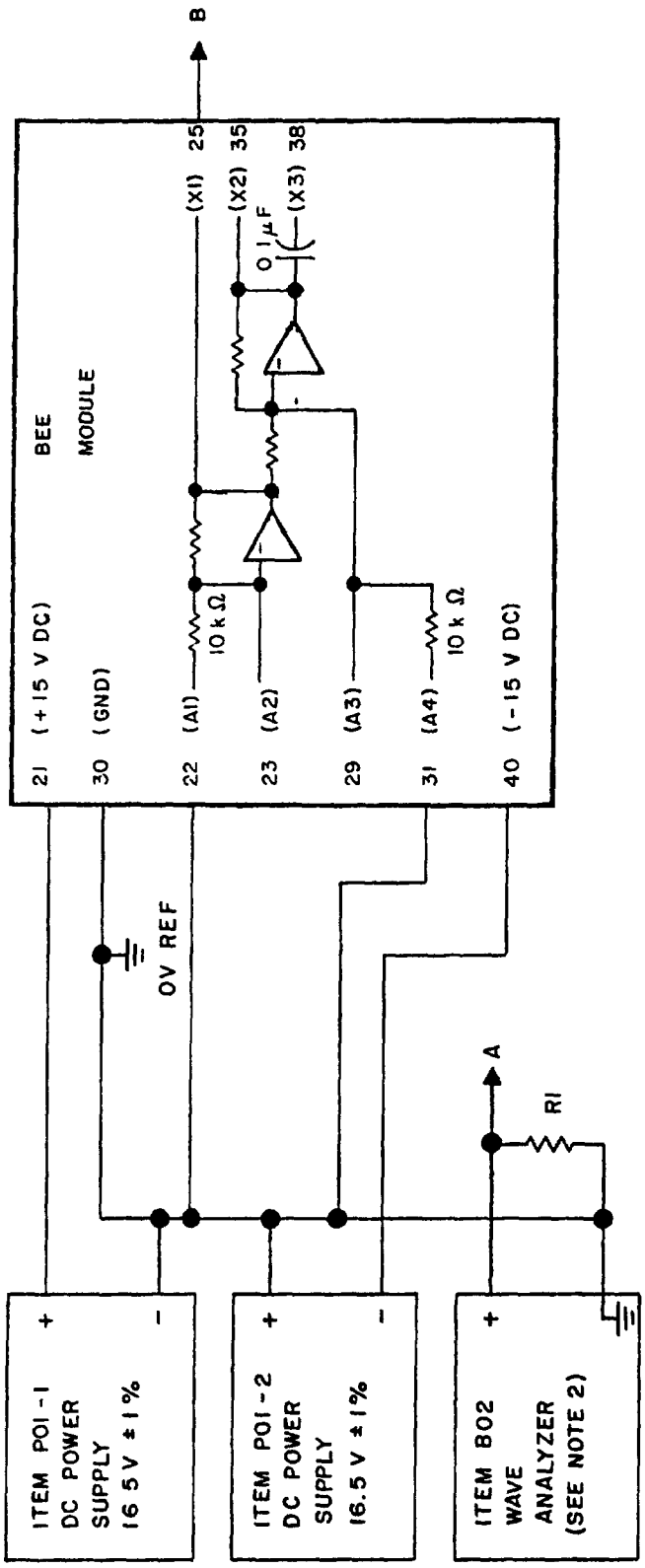
*TABLE VIII. Output noise test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
G1	X1 Output noise	25	Open	0	100	μV rms	11
G2	X2 Output noise	35	30	0	100	μV rms	11

*TABLE IX. Quiescent power supply current test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
H1	V_{CC} Quiescent power supply current	1	20	mA	12
H2	$-V_{CC}$ Quiescent power supply current	1	20	mA	12

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NOTES:
1. RI = 4.99 kΩ ±1%, 1/8 W.
2. Item B02 setup:
   Set bandwidth to 10 Hz.
   Tune from 10 Hz to 20 kHz. The reading should agree with the limits specified in table VIII.

* FIGURE 11. Output noise test circuit for table VIII.

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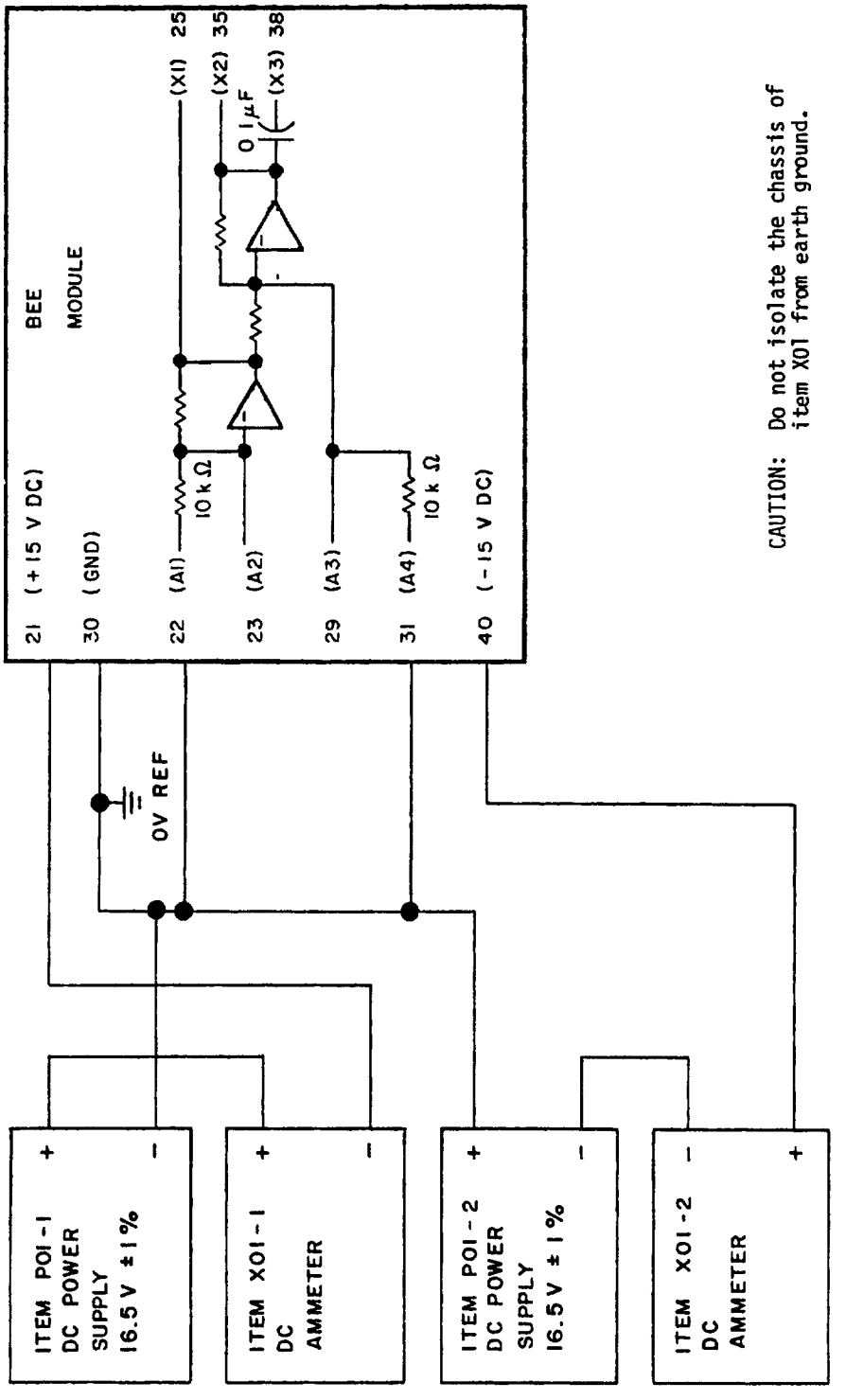


FIGURE 12. Quiescent power supply current test circuit for table IX.

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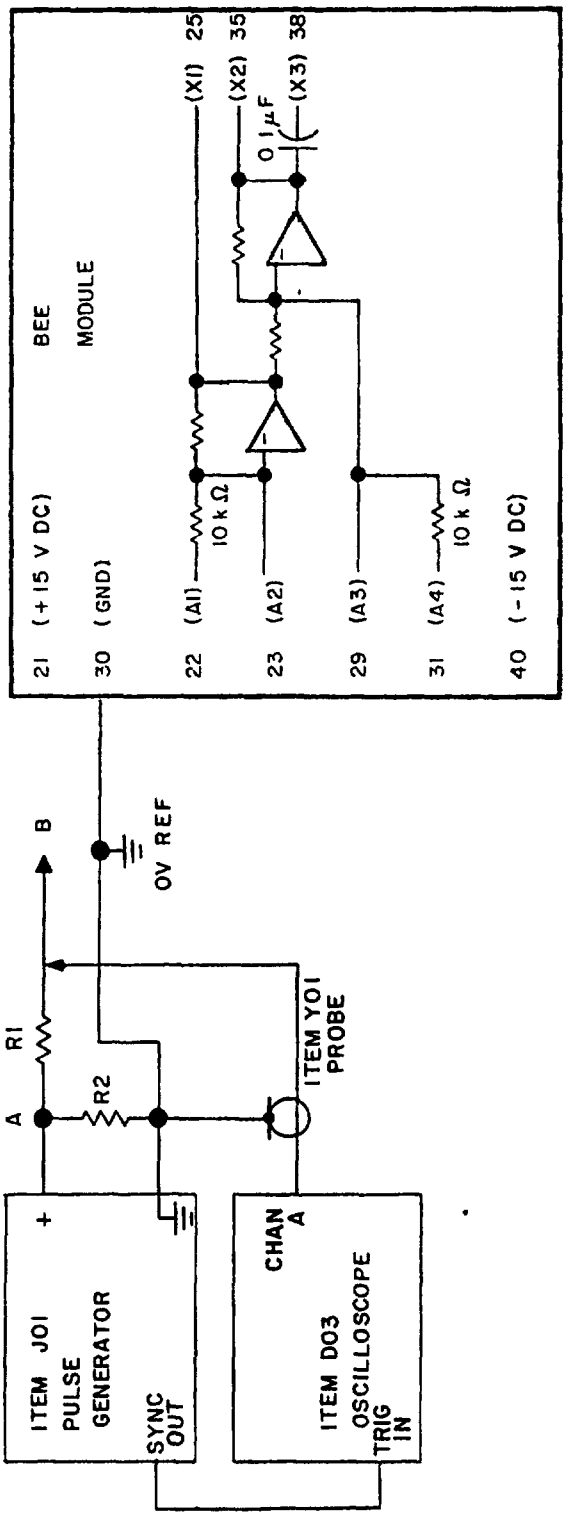
TABLE X. Power supply transient rise time test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		B to pin	Min	Max		
J1	V_{CC} Power supply transient rise time	21	25	75	µs	13,14
J2	$-V_{CC}$ Power supply transient rise time	40	25	75	µs	13,14

TABLE XI. Power supply ripple rejection check test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
K1	X1, V_{CC} Power supply ripple rejection check	25	0.0	1.5	mV rms	15
K2	X1, $-V_{CC}$ "	25	"	"	"	"
K3	X2, V_{CC} "	35	"	"	"	"
K4	X2, $-V_{CC}$ "	35	"	"	"	"

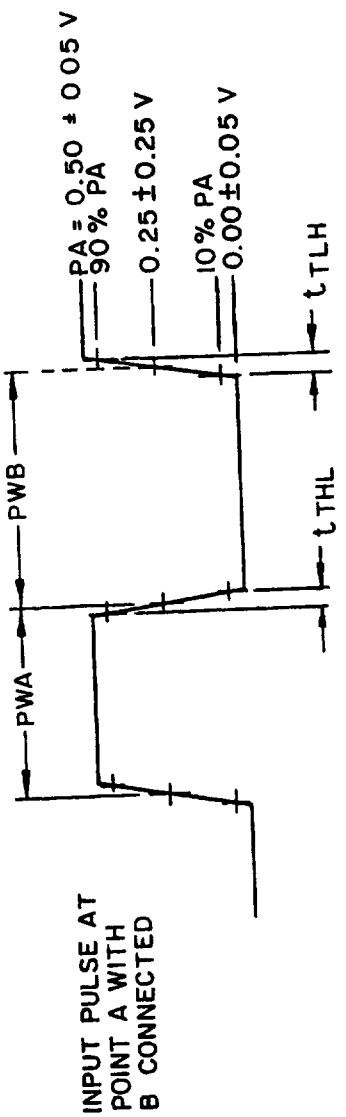
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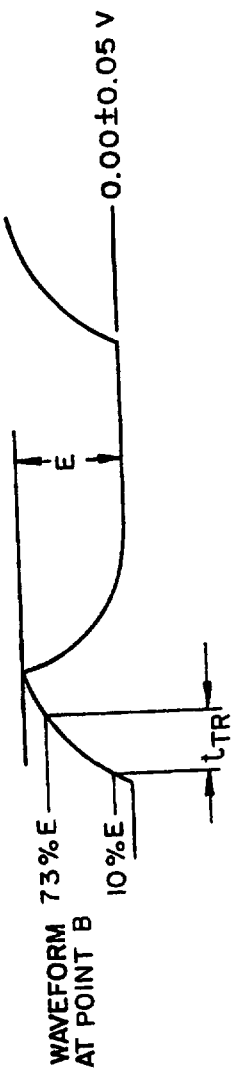


NOTES:
1. R1 = 412Ω ±1%, 1/8 W.
2. R2 = 50Ω ±1%, 1/2 W.

* FIGURE 13. Power supply transient rise time test circuit for table X.

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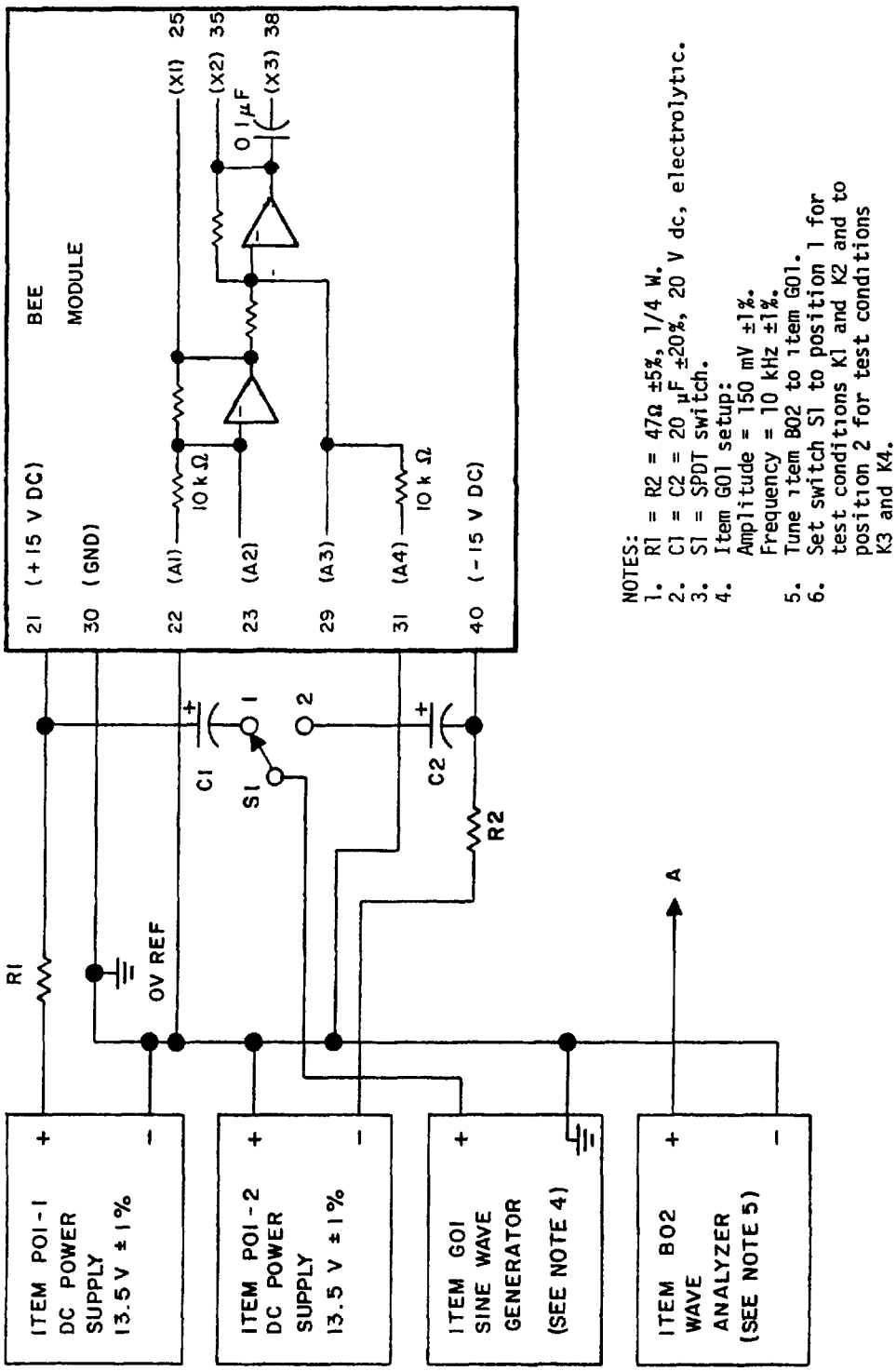




NOTES:
1. Input pulse parameters:
   PWA = PWB = 300 ±30 µs.
   PRP = 600 ±60 µs.
   t_{TLH} and t_{THL} = 500 ns maximum.
2. Measure transient rise time (t_{TR}) at point B.

* FIGURE 14. Power supply transient rise time test circuit waveforms for table X.

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NOTES:
1. R1 = R2 = 47Ω ±5%, 1/4 W.
2. C1 = C2 = 20 μF ±20%, 20 V dc, electrolytic.
3. S1 = SPDT switch.
4. Item G01 setup:
   Amplitude = 150 mV ±1%.
   Frequency = 10 kHz ±1%.
5. Tune item B02 to item G01.
6. Set switch S1 to position 1 for test conditions K1 and K2 and to position 2 for test conditions K3 and K4.

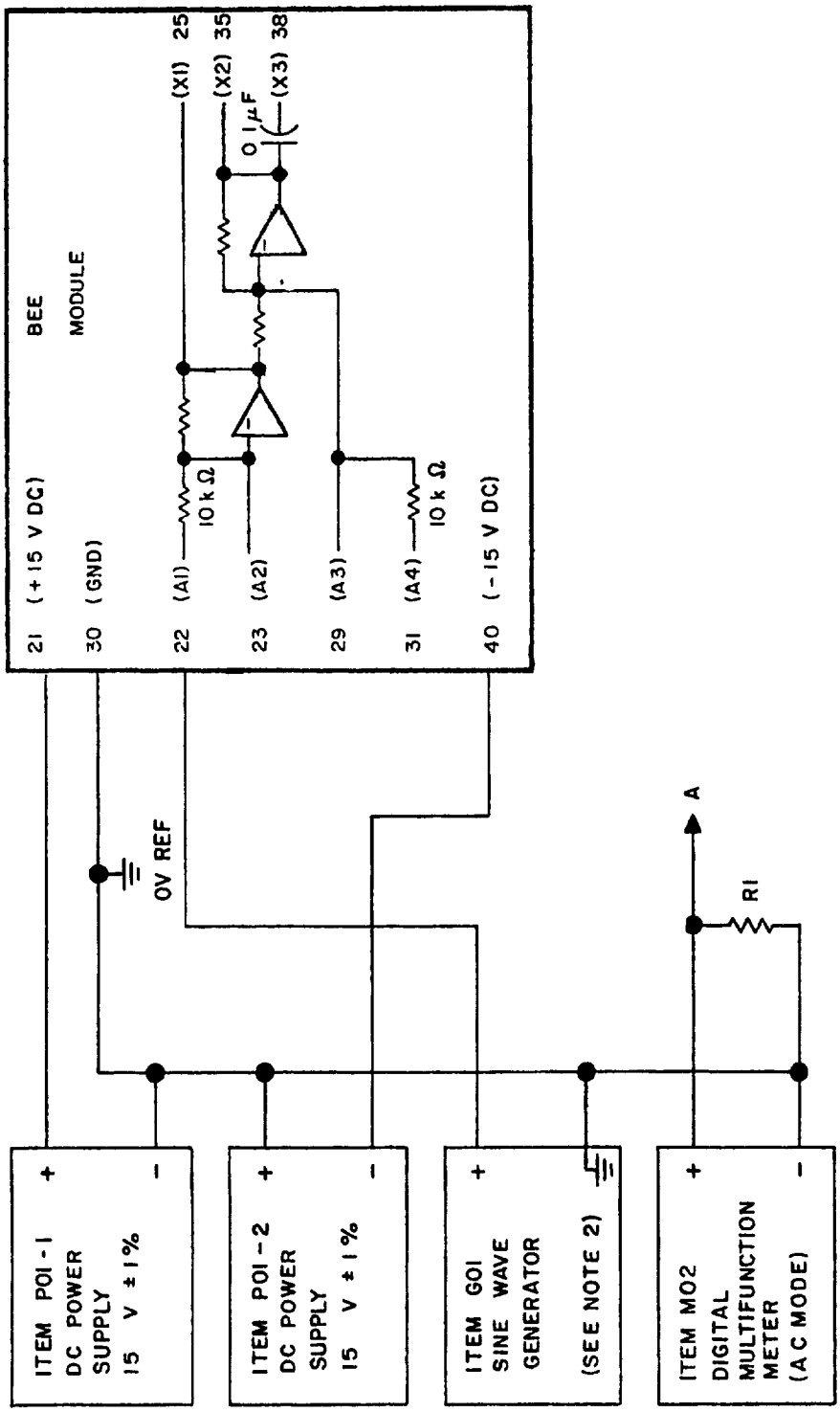
* FIGURE 15. Power supply ripple rejection check test circuit for table XI.

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TABLE XII. Large signal bandwidth check test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
L1	X2 Large signal bandwidth check	35	2.90	3.10	V rms	16
L2	X2 "	35	0	6	"	"
L3	X3 "	38	0.0	3.1	"	"
L4	X3 "	38	0	6	"	"

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NOTES:
1. R1 = 2 kΩ ±1%, 1/8 W.
2. Item G01 setup:
   Initial frequency = 20 kHz ±2% for test conditions L1-L4.
   Final frequency = 100 kHz ±2% for test conditions L1 and L3.
   = 1 MHz ±2% for test conditions L2 and L4.
   For each test condition sweep from initial frequency setting to final frequency setting.
   Amplitude = 3 V rms ±1%.

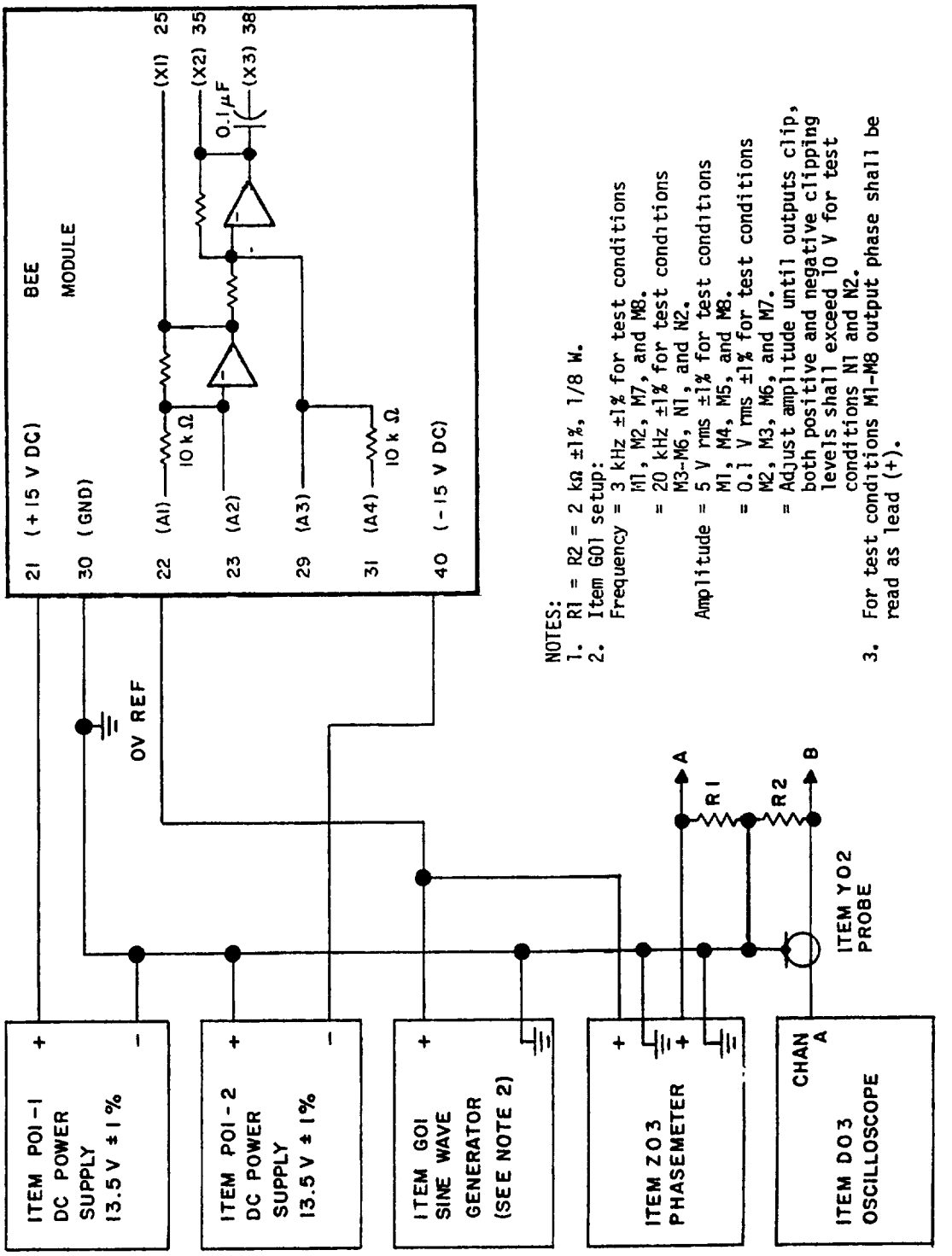
* FIGURE 16. Large signal bandwidth check test circuit for table XII.

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TABLE XIII. Phase shift and output voltage swing test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
M1	X1	Phase shift	25	Open	178	182	Degree	17
M2	X1	"	"	"	"	"	"	"
M3	X1	"	"	"	"	"	"	"
M4	X1	"	"	"	"	"	"	"
M5	X3	"	38	"	0.2	4.2	"	"
M6	X3	"	"	"	0.2	4.2	"	"
M7	X3	"	"	"	12.8	16.8	"	"
M8	X3	"	"	"	12.8	16.8	"	"
N1	X3	Output voltage swing	Open	38	10.00	15.15	Vpk	"
N2	X1	Output voltage swing	Open	25	10.00	15.15	Vpk	"

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NOTES:
1. R1 = R2 = 2 kΩ ±1%, 1/8 W.
2. Item G01 setup:
   Frequency = 3 kHz ±1% for test conditions M1, M2, M7, and M8.
   = 20 kHz ±1% for test conditions M3-M6, N1, and N2.
   Amplitude = 5 V rms ±1% for test conditions M1, M4, M5, and M8.
   = 0.1 V rms ±1% for test conditions M2, M3, M6, and M7.
   = Adjust amplitude until outputs clip, both positive and negative clipping levels shall exceed 10 V for test conditions N1 and N2.
3. For test conditions M1-M8 output phase shall be read as lead (+).

* FIGURE 17. Phase shift and output voltage swing test circuit for table XIII.

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